System Requirements Document
Medicine Dispensing Device

Mechanical Design V2 - Removable Compartment Architecture

Field	**Value**
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Version	1.1
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Supersedes	SRD v1.0 where requirements conflict with the new removable-compartment mechanical design.
Purpose	Define hardware/software architecture, interfaces, states and verifiable requirements for the V2 mechanical concept.

1. System Scope and Context

The system is a single-patient desktop medication dispenser using a rotating carrier with 11 removable dose compartments. Each compartment is magnetically retained on the carrier. A fixed camera station performs medication verification before hand-off. After verification, the carrier moves the target compartment to a fixed access opening where the patient removes the entire compartment. The system later verifies whether removal occurred and records adherence/pickup status.

```
Patient / Caregiver
      |
 Device / Remote UI
      |
      v
   API Layer
      |
      v
 Application Workflows
   |    |    |     |
Schedule Inv. Vision Events/Notify
           |
        Camera
      |
  Device Service
      |
    Serial
      |
 Arduino Firmware
   |    |     |
 Motor Sensors Lighting
```

|
12 indexed carrier stations
= 11 removable compartments
+ 1 fixed hand-off/access station

2. Architecture Requirements

SYS-ARCH-001: The first-generation product shall use a modular monolithic Python application and a separate embedded hardware controller.

SYS-ARCH-002: Application workflows shall coordinate cross-subsystem medication operations; the scheduler shall not directly control hardware.

SYS-ARCH-003: The embedded controller shall own low-level motor, sensor, illumination, homing and interlock behavior.

SYS-ARCH-004: Medication identity/verification decisions shall execute in the application/vision layer, not in embedded firmware.

SYS-ARCH-005: The software shall support fake Device and Camera adapters for end-to-end testing without physical hardware.

3. Mechanical System Requirements

3.1. Carrier and Removable Compartments

HW-CAR-001: The rotating carrier shall provide 12 indexed angular stations, with up to 11 stations occupied by removable medication compartments and one station reserved for the fixed hand-off/access opening.

HW-CAR-002: Nominal angular separation between indexed stations shall be 30 degrees.

HW-CAR-003: Exactly 11 removable dose compartments shall constitute the first-generation loaded capacity.

HW-CAR-004: Each removable compartment shall have a unique human-readable identifier that maps to a persistent software compartment ID.

HW-CAR-005: Each removable compartment shall include a metal/ferromagnetic retention feature positioned to mate with a magnet attached to the rotating carrier inner wall.

HW-CAR-006: Magnetic retention shall prevent unintended compartment displacement during specified normal motion and orientation.

HW-CAR-007: Intentional removal at the hand-off station shall be achievable by the intended patient/caregiver without tools.

HW-CAR-008: The rotating carrier shall include the inner wall, side/separating walls and horizontal base needed to locate removable compartments as represented in Mechanical Design V2.

HW-CAR-009: The fixed device structure shall include the stationary camera/inspection structure and patient hand-off opening.

3.2. Retention and Removal Safety

HW-RET-001: Retention force and compartment geometry shall be validated so normal acceleration/deceleration cannot eject or appreciably shift an installed compartment.

HW-RET-002: The hand-off opening shall permit removal of the presented compartment without requiring access to neighboring compartments or moving mechanisms.

HW-RET-003: The system shall provide a means to determine whether a compartment at the hand-off station is fully installed, fully removed, or in an unsafe/ambiguous intermediate condition.

HW-RET-004: Carousel rotation shall be inhibited while the hand-off compartment is in an unsafe/ambiguous intermediate condition.

HW-RET-005: The product-intent enclosure/guard shall prevent patient access to rotating parts except through the intended compartment-removal interface.

3.3. Motion and Positioning

HW-MOT-001: The carrier shall be driven by a stepper motor or equivalent indexed actuator.

HW-MOT-002: The carrier shall include a physical HOME/reference sensor.

HW-MOT-003: At startup and after position uncertainty, the embedded controller shall home before automatic dose presentation is enabled.

HW-MOT-004: The controller shall maintain a discrete logical station index after successful homing.

HW-MOT-005: The design shall not rely solely on commanded motor step count as proof of safe hand-off position.

HW-MOT-006: Motion profiles shall be selected to avoid dislodging magnetically retained compartments.

HW-MOT-007: A repeatable backlash-control strategy shall be used if gear backlash affects inspection or hand-off alignment.

3.4. Inspection Hardware

HW-VIS-001: The camera shall be part of the stationary device structure and shall image the dedicated inspection position.

HW-VIS-002: The rotating carrier shall place each target compartment repeatably within the camera field of view.

HW-VIS-003: Inspection lighting shall be controlled and repeatable.

HW-VIS-004: Camera focus, field of view and exposure/white-balance settings should be fixed or controlled where supported.

HW-VIS-005: The inspection geometry shall allow medication contents to be observed without removing the compartment from the carrier.

3.5. Hand-Off / Pickup Verification

HW-HOF-001: The hand-off station shall be a fixed patient-accessible opening at which the presented removable compartment can be withdrawn.

HW-HOF-002: Medication shall remain inside the removable compartment throughout hand-off; no gravity chute or pill-drop aperture is required by the V2 architecture.

HW-HOF-003: The system shall determine whether the presented compartment remains installed after the configured pickup interval.

HW-HOF-004: Camera-based presence/absence detection may be used for pickup verification; a dedicated presence sensor may supplement it if needed for reliability.

HW-HOF-005: If pickup state cannot be determined confidently, the system shall report an ambiguous/exception condition rather than assume collection.

4. Embedded Controller Requirements

FW-001: Firmware shall implement HOME, MOVE_TO_INSPECTION, MOVE_TO_HANDOFF, LIGHT_ON, LIGHT_OFF, GET_STATUS and RESET/RECOVER-equivalent commands.

FW-002: Firmware shall generate motor step timing and enforce motion timeouts.

FW-003: Firmware shall read home/reference and hand-off safety/presence sensors that are implemented in the hardware.

FW-004: Firmware shall reject motion commands while a hand-off interlock indicates an unsafe compartment condition.

FW-005: Firmware shall return deterministic ACK, status, DONE or ERROR responses for accepted commands.

FW-006: Firmware shall enter a non-moving FAULT state when physical position or interlock state is unsafe or unknown.

FW-007: Firmware shall not decide medication-verification success or patient adherence status.

4.1. Embedded Hardware State Model

State	Meaning	Allowed next states
POWER_ON	Controller initialized; motion inhibited until reference established.	HOMING, FAULT
HOMING	Finding physical reference.	READY, FAULT
READY	Known station index and safe hand-off state.	MOVING, FAULT
MOVING	Carrier rotation active.	READY, FAULT
HANDOFF_WAIT	Compartment presented; motion inhibited pending safe removal/presence state.	READY, FAULT
FAULT	Unsafe or unknown physical condition.	RECOVERY/HOMING after explicit action

4.2. Serial Interface

IF-SER-001: Host and embedded controller shall use a documented serial request/response protocol with command IDs.

IF-SER-002: Protocol shall define framing, command parameters, station/compartment identifiers, status/error codes, timeout and reconnect behavior.

IF-SER-003: The host shall issue device-level commands and shall not generate individual step pulses.

```
CMD,104,MOVE_TO_INSPECTION,COMPARTMENT=3
DONE,104,STATION=INSPECTION,COMPARTMENT=3
CMD,105,MOVE_TO_HANDOFF,COMPARTMENT=3
ERROR,105,HANDOFF_INTERLOCK_UNSAFE
```

5. Software System Requirements

5.1. Platform and Boundaries

SW-PLT-001: The application backend shall be implemented in Python.

SW-PLT-002: SQLite shall be the local first-generation persistent store.

SW-PLT-003: The UI shall consume a validated local API; FastAPI is recommended for new production implementation.

SW-PLT-004: The browser UI should use React/TypeScript or equivalent modern frontend technology.

SW-PLT-005: Serial hardware access shall be isolated behind a Device Service.

SW-PLT-006: Production code shall not depend on prototype package namespaces or prototype data paths.

5.2. Domain Model

Entity	Role
Patient	Identity, timezone, active state.
Medication	Canonical medication identity.
Schedule	Recurring/one-time medication schedule.
ScheduledDose	Concrete dose occurrence.
RemovableCompartment	Physical compartment ID, carrier station, presence and lifecycle state.
CompartmentMedication	Expected/loaded medication and quantity.
InspectionEvent	Medication-content verification or compartment-presence inspection.
PresentationEvent	Record that a verified compartment was moved to the hand-off station.
PickupEvent	Collected, not-collected, ambiguous or manually resolved result.
Alert / Notification	User-facing exception and delivery state.
SystemEvent	Auditable user/software/hardware event history.

5.3. Compartment Lifecycle State

State	Meaning
AVAILABLE_EMPTY	Compartment installed and available for loading.
LOADED	Compartment installed with expected medication.
VERIFIED	Loaded contents passed verification.
DUE	Assigned scheduled dose is due.
PRESENTED	Compartment is at the hand-off opening for pickup.

State	Meaning
REMOVED	Compartment has been collected from the device.
RETURNED	Previously removed compartment has been reinstalled and is awaiting load/verification.
ERROR	Presence, identity or mechanical state requires resolution.

SW-CMP-001: The system shall maintain persistent state for all 11 removable compartment identities.

SW-CMP-002: Software logical identity shall not be derived only from current angular position; compartment identity and carrier station shall be separate fields.

SW-CMP-003: A removed compartment shall remain unavailable for future scheduled assignment until it is reinstalled and returned to an allowed lifecycle state.

SW-CMP-004: The system shall detect and alert when a compartment expected to be installed is missing before its scheduled dose becomes due.

5.4. Scheduling Requirements

SW-SCH-001: Schedules shall support one-time and recurring patterns with timezone-aware evaluation.

SW-SCH-002: A schedule shall reference medication and quantity; physical removable-compartment assignment shall remain a separate concern.

SW-SCH-003: The scheduler shall identify due doses but shall not directly command motion hardware.

SW-SCH-004: The assignment service shall not assign a dose to a compartment that is REMOVED, ERROR or otherwise unavailable.

SW-SCH-005: Mid-cycle schedule changes shall not silently alter already-loaded compartment contents.

5.5. Vision Requirements

SW-VIS-001: The Vision Service shall support pre-hand-off medication-content verification.

SW-VIS-002: The Vision Service shall return detected medication/pill identity, count, confidence and supporting geometry/annotation data where available.

SW-VIS-003: The application shall compare expected versus detected contents and configured acceptance thresholds.

SW-VIS-004: Low-confidence or mismatched medication verification shall block presentation at the hand-off station unless an authorized override is recorded.

SW-VIS-005: The Vision Service shall support compartment presence/absence classification at the hand-off station when camera-based pickup verification is used.

SW-VIS-006: Ambiguous presence/absence results shall not be automatically converted to COLLECTED.

5.6. Persistence and Events

SW-DB-001: SQLite foreign-key enforcement shall be enabled.

SW-DB-002: Compartment identity, carrier position and current presence state shall be persistently stored.

SW-DB-003: Schedule information shall be stored separately from compartment assignment and physical presence.

SW-DB-004: Related state transitions shall be committed transactionally where partial persistence would create inconsistency.

SW-LOG-001: Event history shall record load/reinstall, pre-dose inspection, presentation, pickup verification, timeout/not-collected, override, homing and hardware faults.

6. Integrated Operational Workflows

6.1. Scheduled Presentation Workflow

1. Scheduler identifies a ScheduledDose as due.
2. Application confirms the assigned removable compartment is installed, loaded and eligible.
3. Application confirms device READY state and safe hand-off condition.
4. Device Service moves the target compartment to INSPECTION.
5. Firmware confirms position.
6. Application turns on controlled lighting, captures an image and invokes medication verification.
7. If verification fails, application records BLOCKED and creates an alert; compartment is not presented.
8. If verification passes, application commands MOVE_TO_HANDOFF for the same physical compartment.
9. Firmware confirms hand-off position and enters HANDOFF_WAIT or equivalent safe state.
10. Application records PRESENTED and notifies the patient.
11. After the configured pickup interval, application acquires compartment-presence evidence.
12. If removal is confirmed, application records COLLECTED/REMOVED and updates compartment availability.
13. If the compartment remains present, application records NOT_COLLECTED/MISSED according to policy and notifies the patient/caregiver.
14. If presence is ambiguous, application records an exception and requests resolution without assuming pickup.

6.2. Product-Level Dose State Machine

State	Meaning
SCHEDULED	Dose exists and awaits due time.
PREPARING	Application validates compartment and hardware preconditions.
POSITIONING_INSPECTION	Carrier moves target compartment to camera station.
INSPECTING	Medication-content verification active.
VERIFIED	Expected contents passed verification.
POSITIONING_HANDOFF	Carrier moves same compartment to access opening.
READY_FOR_PICKUP	Compartment is presented and motion is inhibited as required.
COLLECTED	Removal of the compartment is confirmed.
NOT_COLLECTED	Pickup window expired with compartment still present.

State	Meaning
AMBIGUOUS_PICKUP	Presence/removal could not be determined confidently.
BLOCKED	Verification or safety condition prevents presentation.
FAILED	Hardware/software operation failed safely.

6.3. Return / Refill Workflow

1. Caregiver identifies removed compartments requiring return/refill.
2. Caregiver reinstalls a compartment into an allowed carrier position.
3. System verifies presence and associates physical compartment ID with the expected logical identity/position.
4. Caregiver loads the scheduled medication.
5. System performs optional/required visual verification.
6. Compartment becomes AVAILABLE/VERIFIED for future assignment.

6.4. Startup and Recovery

SYS-REC-001: After startup, the system shall not trust the previously persisted angular carrier position without re-establishing physical reference.

SYS-REC-002: The embedded controller shall home before automatic scheduled presentation resumes.

SYS-REC-003: The application shall reconcile hardware position with persisted compartment-presence state.

SYS-REC-004: If a compartment that was PRESENTED before power loss is missing after restart, the system shall record a recovery exception and require policy-based resolution rather than infer an exact pickup time.

SYS-REC-005: Loss of serial communication shall place the application into a non-moving/non-presenting state until hardware status is re-established.

7. User Interface Requirements

UI-001: Dashboard shall show next dose, current device state, active alerts and count of installed/available removable compartments.

UI-002: Carousel view shall display all 11 compartment IDs, their logical states and the fixed inspection/hand-off station.

UI-003: Compartment detail shall show current presence, expected/loaded medication, latest inspection and pickup history.

UI-004: Guided loading/refill shall identify compartments that are removed, returned, awaiting load or awaiting verification.

UI-005: The local UI shall clearly indicate READY_FOR_PICKUP and the compartment ID presented to the patient.

UI-006: The UI shall show NOT_COLLECTED and AMBIGUOUS_PICKUP exceptions and provide caregiver resolution actions.

UI-007: Development mode shall support fake compartment removal and fake hardware adapters.

8. Safety and Non-Functional Requirements

NFR-SAF-001: Unknown carrier position, unsafe hand-off interlock, failed homing or failed required medication verification shall prevent automatic presentation.

NFR-SAF-002: The embedded controller shall reject movement if a presented compartment is partially removed or the hand-off station is otherwise unsafe.

NFR-SAF-003: Hardware faults shall default to a non-moving state.

NFR-SAF-004: The device shall not treat PRESENTED as COLLECTED without positive removal evidence or explicit authorized resolution.

NFR-TST-001: Motion, retention, presence detection, vision verification and state transitions shall be testable with repeatable automated or bench procedures.

NFR-TST-002: Software workflows shall be executable with fake Device and Camera adapters.

NFR-MNT-001: Configuration such as serial port, vision thresholds, pickup interval, motor timing and camera settings shall be externalized.

9. Interface Control Summary

Interface	Producer	Consumer	Data / commands
HTTP/API	Browser UI	Python API	Medication, schedules, compartment states, presentation/pickup actions, alerts.
Workflow -> Scheduler	Application	Scheduler	Due-dose query and schedule validation.
Workflow -> Compartment Service	Application	Inventory/Compartment Service	Identity, presence, station, expected/loaded contents.
Workflow -> Vision	Application	Vision Service	Inspection image + expected contents or presence-classification request.
Workflow -> Device	Device Service	Arduino	Home, move-to-inspection, move-to-handoff, light, status, recover.
Arduino -> Host	Arduino	Device Service	ACK/status/DONE/error, station index, interlock/presence state.
Camera -> Vision	Camera Adapter	Vision Service	Captured image.
Sensors -> Firmware	Home/hand-off sensors	Arduino	Physical reference and safe-presence state.

10. Verification and Acceptance Plan

Verification area	Method	Pass evidence
Magnetic retention	Repeated loaded rotation tests	No unintended displacement/ejection of any installed compartment.
Manual removal	Human-factor bench test	Intended user can remove presented compartment without tools or disturbing adjacent compartments.
12-station indexing	Repeated homing/indexing test	All compartment IDs repeatedly reach inspection and hand-off alignment.
Inspection framing	Image repeatability test	Each installed compartment is consistently framed for medication analysis.
Pickup detection	Presence/absence test	System reliably distinguishes installed versus removed compartment.
Partial-removal interlock	Fault-injection test	Motion is rejected while compartment is unsafe/partially removed.
Power-loss recovery	Power interruption test	System re-homes and reconciles compartment presence without unsafe movement.
End-to-end workflow	Integrated scenario	Due -> inspect -> verify -> hand-off -> removal verification -> DB/UI update.

11. Requirements Removed or Replaced from SRD v1.0

SRD v1.0 concept	V1.1 disposition
Stationary floor and bottom dispense aperture	Removed; no loose-pill gravity release in Mechanical Design V2.
Chute and drop sensor	Removed/replaced by removable-compartment hand-off and pickup verification.
Receiving bucket	Removed; patient receives the whole compartment.
DISPENSING hardware state	Replaced by HANDOFF_WAIT/presentation control.
DISPENSED -> AWAITING_ACKNOWLEDGEMENT -> TAKEN	Replaced by PRESENTED -> READY_FOR_PICKUP -> COLLECTED / NOT_COLLECTED.
12 loaded compartments	Revised to 11 removable dose compartments plus one fixed access/hand-off station.

12. Open Engineering Decisions

Decision	Evidence needed
Magnet type/retention force	Loaded rotation and manual-removal force testing across intended users.
Compartment identity sensing	Determine whether human-readable IDs are sufficient or machine-readable tags/markers are required.
Hand-off presence sensing	Compare camera-only detection with switch/Hall/optical sensor supplementation.
Partial-removal interlock mechanism	Mechanical/electrical method must robustly inhibit motion.
Camera field of view	Verify one camera can support medication inspection and hand-off presence inspection, or define a second sensor/camera.
Carrier bearing/drive geometry	Repeatability, torque, backlash and noise testing with all 11 compartments installed.
Return/reinstall workflow	Validate caregiver ergonomics and error-proofing for placing each identified compartment.